USN. | | | | | | | | | | [24MTAS102]

FIRST SEMESTER M.Tech EXAMINATION, JANUARY- 2025

INTRODUCTION TO AEROSPACE ENGINEERING

Time: 3 Hours **Maximum Marks: 100**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 20 marks.
ii. Missing data may be suitably assumed.

		Questions	CO's	Blooms Level	Marks
1.	a.	i. Describe the properties of the five layers of the earth's atmosphere with a neat diagram. ii. Explain the international standard atmosphere (ISA)	1	L2	5 5
	b.	i. Discuss the functioning of the flight controls in a helicopter. ii. How does the swash plate mechanism in a helicopter work? Explain in brief with a neat diagram.	1	L2	6 4
2.	a.	Define the following properties of materials: Hardness, Strength, Malleability, Ductility, Elasticity, Toughness, Conductivity and Thermal Expansion.	2	L2	10
	b.	i. What are composites? Explain their advantages. ii. Describe the construction of V-n diagram.	2	L2	4 6
3.	a.	i. Draw the T-s diagram for an ideal Brayton cycle and explain the process in the machine where it is practically applied. ii. Discuss the working of the ramjet engines along with their advantages and disadvantages.	3	L2	6 4
	b.	i. Write the full equation for the thrust produced by a turbojet engine annotating the meaning of each term in the equation. Thrust (N) = m_e x Ve - m_a x Vo + Ae x (Pe - Po) - Ai x (P_i - Po) ii. The exit velocity from the jet unit is 650 m/s for an air flow of 46 kg/s through unit. The aircraft is flying at 250 km/hr. Calculate thrust developed, thrust power and propulsive efficiency.	3	L2 L3	5 5
		OR			
4.	a.	i) Draw the internal combustion engine family tree. ii) Explain how the pressure, velocity and temperature varies in the gas flow through a gas turbine engine with a neat diagram.	3	L2	5 5

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	b.	i. Draw the T-s diagram for the cycle applicable to turbo fan engine. ii. Explain turbofan (mixing and non-mixing exhaust flow) engines with neat diagrams.	3	L2 L2	5 5
5.	a.	i. Explain in brief the working of a typical hydraulic system. ii. Describe the various hydraulic pumps used in aircrafts.	4	L2 L2	4 6
	b.	Explain the types of fuel level sensors used in aircraft.	5	1	10
		OR			
6.	a.	Explain in detail the working of the flight instruments based on the pitot-static system.	4	L2	10
	b.	Explain in brief the functioning of the following: i. Instrument Landing System (ILS) with Marker Beacons. ii. Automatic Direction Finders (ADF)	5	L2	6 4
7.	a.	i. State Keppler's laws and explain their importance. ii. Explain the different orbits of satellites orbiting earth.	6 6	L2	7 3
	b.	Calculate the escape velocity of an object from earth.	6	L2	10
		OR			
8.	a.	i. Describe in brief the different type of chemical propellants used in rocket motors. ii. Explain the differences between tactical and strategic missiles.	6	L2	7 3
	b.	i. Describe the different types of launch vehicles and their purposes. Name the various launch vehicles developed by ISRO and their use in brief. ii. Sketch and explain the different types of typical solid propellant grains.	6	L2	7 3